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PALEOMAGNETISM OF LATE PRECAMBRIAN TO EARLY PALEOZOIC MIXTITE-BEARING FORMATIONS IN NAMIBIA (SOUTH WEST AFRICA): THE NAMA GROUP AND BLAUBEKER FORMATION†

A. KRÖNER,* M. O. McWILLIAMS,** G. J. B. GERMS,***
A. B. REID,**** and K. E. L. SCHALK*****

ABSTRACT. Paleomagnetic results from two mixtite-bearing sedimentary sequences of Namibia (South West Africa) are reported, the late Precambrian Blaubeker Formation of the Nosib Group and the late Precambrian to early Paleozoic Nama Group. Units from these sequences record complex magnetizations which are a product of metamorphic overprinting during the Pan African tectonothermal activity. Three sites from the Blaubeker Formation appear to have acquired a magnetization of post-folding age (NBX; D=345, I=+23, α_{95}=20°) probably as a result of the Damaran phase of Pan-African orogenesis. The lower Nama (Kuibis and Schwarzrand Subgroups) records a probable primary remanence (N1; D=020, I=−11, α_{95}=13°) of late Precambrian age and a secondary remanence (N3; D=110, I=−65, α_{95}=10°) of early Paleozoic age. The upper Nama (Fish River Subgroup) displays a probable pre-folding magnetization (N2; D=087, I=+31, α_{95}=15°) of early Cambrian age, intermediate between N1 and N3 on the African apparent polar wander path. Some sites in the Fish River Subgroup also record the N3 overprint magnetization observed in the lower Nama. In view of the significant polar difference between N1 and N2 we infer a considerable hiatus between the upper and lower Nama and suggest that the Cambrian-Precambrian boundary lies at the unconformity separating these units. The Blaubeker Formation records two apparent magnetizations, both of which are probably related to Pan-African thermal events in the Damara orogenic belt. Low depositional paleolatitudes for the lower Nama are in apparent conflict with glacial phenomena in these rocks. Present data are insufficient to decide whether these features were caused by a circumpolar or a low latitude glaciation.

INTRODUCTION AND STATEMENT OF PROBLEM

Many geologists have been intrigued by the great number of apparently glaciogenic sedimentary rocks reported from the upper Proterozoic of southern Africa (Schwarzbach, 1961, 1974; Kröner and Rankama, 1973), and there is a clear need for an unbiased reassessment of these strata as concerns their depositional environment (Schermerhorn, 1974, 1975; Kröner and Rankama, 1973). Paleomagnetism is the most important tool for providing unequivocal information on the paleolatitudes at which such rocks were laid down, and a comprehensive project has therefore been started to determine the paleomagnetic pole positions for a variety of late Precambrian to early Paleozoic sedimentary units in southern Africa and Australia that contain proven or possible glaciogenic strata.

We report here the results from two mixtite-bearing lithostratigraphic units of Namibia (South West Africa): the late Precambrian Blaubeker Formation of the Nosib Group and the overlying late Pre-

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* Institut für Geowissenschaften, Johannes Gutenberg-Universität, Mainz, Federal Republic of Germany

** Research School of Earth Sciences, Australian National University, Canberra, A.C.T. Australia. Present address: Geophysics Department, Stanford University, Stanford, California 94305

*** Namibia (South West Africa) Geological Survey, Windhoek, Namibia (South West Africa). Present address: J. C. I. Exploration Research Unit, Randfontein, South Africa

**** Department of Physics, University of Zimbabwe (Rhodesia), Salisbury, Zimbabwe (Rhodesia)

***** Namibia (South West Africa) Geological Survey, Windhoek, Namibia (South West Africa)

cambrian to early Paleozoic Nama Group. Both units contain rocks that display features that, on the basis of commonly used criteria (Flint, 1971, 1975; Frakes and Crowell, 1967), are characteristic of the presence of ice during deposition (Kröner and Rankama, 1973). The data from other late Precambrian mixtites of southern Africa and Australia are reported elsewhere (McWilliams, ms; McWilliams and Kröner, 1980; McWilliams and McElhinny, 1980).

An important aspect of paleomagnetism is its usefulness in stratigraphic correlation, in assessing the importance of depositional breaks, and in determining the age of otherwise largely undatable rock sequences by comparing respective paleomagnetic poles with a previously established apparent polar wander path (APWP). This is of particular interest for the Nama Group whose lower part contains a still undated but presumed late Precambrian fauna (Germs, 1972a,b), whereas the disconformably or paraconformably overlying upper part may be Cambrian in age (Germs, 1972a; B. Daily, personal commun., 1977). The early Paleozoic APWP for Africa is reasonably well established (Schmidt and Morris, 1977; Morris and Carmichael, 1978); thus it was our intention to discover whether the Nama Group is entirely Precambrian or only partly so.

REGIONAL DISTRIBUTION, STRATIGRAPHY AND AGE

Blaubeker Formation.—The Blaubeker Formation outcrops at four localities in the southern foreland of the Damara eugeosyncline (fig. 1). It rests unconformably upon the Kamtsas Formation of the Nosib Group and is overlain unconformably, or locally pseudoconformably, by the basal portions of the Nama Group (Schalk, 1972).

The Blaubeker succession comprises conglomerate, mixtite, quartzite, and minor shale, altogether several hundred meters thick. Samples were collected on the farms Blaubeker 488, Lepel 339, and Garies Oos 489 (Rehoboth District) from quartzite intercalated in the basal conglomeratic portion of the formation. Their stratigraphic position is indicated in the schematic section of figure 2. The mixtite, a prominent member of the succession, is about 200 m thick on Blaubeker 488 but apparently considerably thicker farther to the east. It contains abundant striated pebbles and blocks as well as "dreikanters" which indicate accumulation under euglacial or periglacial conditions. Most of these inclusions seem to be redeposited pebbles originally contained in the underlying Kamtsas quartzite which became striated during transport (Schalk, 1972). The upper portion of the sequence consists of conglomeratic feldspathic quartzite. The patchy distribution of the Blaubeker Formation suggests that the formation was originally deposited over an extensive area but was largely removed by subsequent pre-Nama erosion.

Because of the similarity of many of the Blaubeker rocks to those of the underlying Kamtsas Formation, the formation is regarded as the upper portion of the Nosib Group; it is probably a correlative of tillitic rocks occurring in the Nosib Group to the south of Tsumeb and of the Numees mixtite of the Gariep Group in the Orange River area. The age of the Blaubeker Formation is thus not known but is probably in the range 840 to 900 m.y. (Kröner, 1977).

Although no major structural deformation is apparent in the area where the Blaubeker Formation was sampled, dips of 10 to 50 degrees are common, and incipient cleavage and recrystallization in these rocks indicate that a low grade regional metamorphism (anchimetamorphism) has occurred. The metamorphism could be a result of either the early (580-680 m.y.) or late (450-570 m.y.: Damaran) episodes of the Pan-African orogeny (Clifford, 1967). Radiometric whole-rock and mineral ages from metamorphic and plutonic rocks near the sampling area have a mean value of 471±26 m.y. (± 1 standard deviation: Rb–Sr, K–Ar, and U–Pb ages of Clifford, 1967; Burger and Coertze, 1973; Kröner and Hawkesworth, 1977; Haack and Hoffer, 1976), suggesting that the

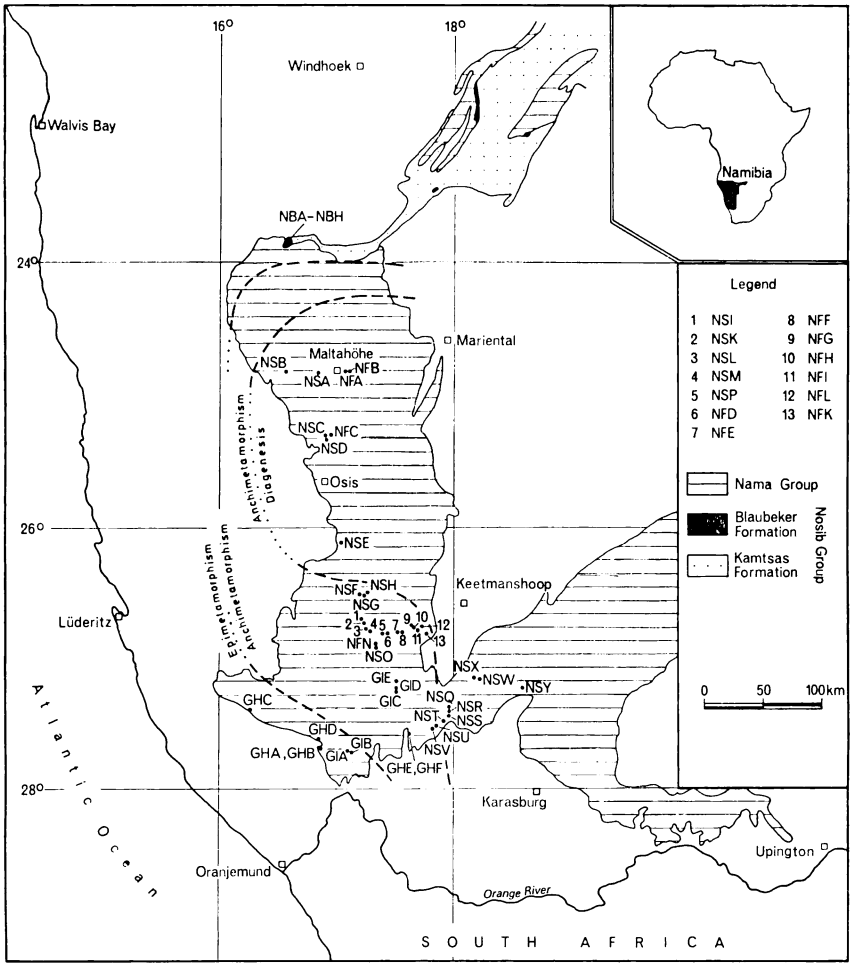


Fig. 1. Map of southern Nambia showing extent of Nama Group and Blaubeker Formation outcrops together with sampling sites. Dashed and dotted lines delineate approximate isograds for very low grade metamorphism.

Damaran episode might have been the most recent heating event to have affected the Blaubeker samples.

Nama Group.—The beds of the Nama Group cover an extensive area in the southern part of Namibia and the adjoining parts of the northwestern Cape Province of South Africa (fig. 1). The Nama Group sediments were deposited on a predominantly gneissic, stable continental platform (the Kalahari Platform) and have undergone only very localized deformation and metamorphism during Pan-African tectonism (Germs, 1974). Very low grade metamorphism has affected the sedimentary rocks along their western and northern outcrop margin (Germs, 1972a; Ahrendt, Hunziker, and Weber, 1978). The beds are predominantly flat lying or have been slightly folded into broad synclinoria and monoclinoria during the Damaran (450-570 m.y.) episode with dips generally less than 10 degrees. Paleomagnetic sampling was confined to unmetamorphosed or weakly metamorphosed regions only. The stratigraphy and paleontology of the Nama Group have been studied in detail by Germs (1972a). A schematic lithostratigraphic subdivision is given in table 1.

Each of the three Nama units generally comprises a very characteristic lithologic entity. The basal Kuibis Subgroup consists largely of detrital quartzitic sediments, dolomitic limestone, and minor shale, while the Schwarzrand Subgroup contains greenish sandstone, siltstone, thick shale, and dark blue limestone. The overlying Fish River Subgroup is characterized by predominantly red sandstone and shale. A roughly east-west trending basement ridge at Osis (fig. 1) was responsible for the development of two contrasting facies during Kuibis and lower to middle Schwarzrand deposition (table 1).

The occurrence of intraformational conglomerates and mixtites as well as grooved and scraped bedding planes provide evidence for two possible cold climatic periods which may have been accompanied by local glaciations during deposition of the Schwarzrand Subgroup (Germs, 1974; Schwellnus, 1942; Kröner and Germs, 1971). Similar features indicative of a cold depositional environment have been reported from some beds of the Fish River Subgroup (Hälbich, 1964).

The precise ages of the different Nama units are still uncertain. Germs (1974) assigned the succession below the upper Schwarzrand Sub-

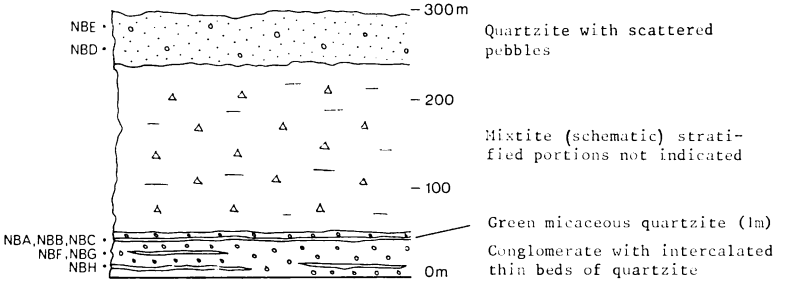


Fig. 2. Generalized section of the Blaubeker Formation, showing stratigraphic distribution of sampling sites.

TABLE 1

Stratigraphy, approximate thickness, and lithology of the various units of the Nama Group west of the Klein Karas Mountains and south of Osis in South West Africa

Subgroup	Formation	Member	Approximate thickness	Lithology
Fish River	Gross Aub	Deurstamp	0- 30 m	red shale and red sandstone
		Rosenhof	0-480 m	red shale including red sandstone, locally greenish
	Nababis	Haribes	0-400 m	red sandstone with minor red shale, locally greenish
		Zamnarib	0- 80 m	red shale and red sandstone
	Breckhorn		10- 80 m	red quartzitic sandstone and some red shale
	Stockdale	Wasserfall	130-240 m	red friable sandstone and minor red shale
		Inachab	11- 70 m	gray to reddish quartzitic sandstone and red shale, with thin basal conglomerate south of Maltahöhe
		Haseweb	0-220 m	red friable sandstone and red shale
		Kabib	0- 10 m	gray to reddish quartzitic sandstone with thin conglomerate at base
Schwarzrand	Nomtsas		0- 60 m	green shale and green sandstone with a basal coarse conglomerate at many places, limestone toward the southwest
	Totem	Spitskop	0-400 m	dark-blue limestone with some green sandstone
		Feldschuhhorn	0-170 m	green shale with minor green sandstone
		Huns	0-240 m	dark-blue limestone with minor intercalated green sandstone and green shale
	Nasep		0-310 m	green sandstone and white quartzite with green shale and black limestone toward the southwest
	Nudaus	Vingerbreek	15-275 m	green shale with minor intercalated greenish sandstone and some blue limestone toward the southwest
		Niederhagen	0- 8 m	gray to greenish quartzite with intercalated green shale
Kuibis		Urikos	0- 14 m	bluish-green shale with interbeds of sandstone, carbonate, and minor shale
	Schwarzkalk		0- 50 m	blue-black limestone with very little green shale
	Dabis	Kliphoek	0-100 m	gray to white quartzite with thin pebbly quartzite locally at base
		Mara	0- 40 m	gray to greenish shale with sandstone replaced by gray dolomitic limestone toward southwest
		Kanies	0-100 m	gray to greenish (quartzitic) sandstone with thin conglomerate or pebbly sandstone at base

group to the age range 600 to 700 m.y., since this unit contains an Ediacara-type fauna (Glaessner, 1971). Pflug (1974 and personal commun., 1978) attributes a lower to middle Vendian age (~600-670 m.y.b.p.) to the Kuibis Subgroup, based on the presence of Metazoan-like petalo-organisms in these rocks. The upper Nama could be very late Precambrian in age (Germs, 1974), although the presence of the trace fossil *Phycodes pedum* may indicate a Cambrian age (Seilacher, 1955; Daily, 1973). South of Witputs (27.5° S) basal Nama rocks rest unconformably on a sequence that contains acid volcanic rocks dated at 686±32 m.y., whereas farther east the Nama is intruded by granitoid rocks of the Bremen Complex, dated at 553±13 m.y. (Rb–Sr whole rock isochron ages using Rb decay constant 1.42; Allsopp and others, 1979). Thus paleontological and geochronological data are compatible with an age range of 550 to 700 m.y. for the Nama. Rb–Sr dating of illite fractions from Kuibis Subgroup shales is currently in progress; preliminary results confirm a late Precambrian age for these rocks (Clauer and Kröner, unpub. data). A schematic section through the entire Nama Group showing the stratigraphic position of sampling sites in the present study is given in figure 3.

PREVIOUS PALEOMAGNETIC WORK

Piper (1975) reported paleomagnetic results from red sandstones at 9 sites in the Fish River Subgroup. His results were based upon site mean directions calculated after alternating field (AF) demagnetization in peak fields up to 160 mT. After cleaning, sample mean directions appeared to define a streaked distribution from a PEF (Present Earth Field) direction to an east-southeast steep negative direction (fig. 13b of Piper, 1975). No thermal demagnetization experiments were attempted. Despite the facts that the overall mean magnetization direction was not significantly different from the PEF direction and that no fold test could be employed, Piper concluded that the observed directions were primary and record a period of rapid apparent polar motion as suggested by McElhinny, Giddings, and Embleton (1974). As is shown below, two components of magnetization have been isolated in the Fish River Subgroup during progressive thermal demagnetization, suggesting that the previous interpretation of a magnetization acquired penecontemporaneously with deposition may have been incorrect. No other paleomagnetic results have been reported to date from the Nama Group or Blaubeker Formation.

PALEOMAGNETIC RESULTS

Samples from the Blaubeker Formation and Nama Group were obtained using portable diamond drilling equipment. Orientation of samples at the outcrop was done by solar compass and clinometer. Generally three cylindrical specimens of nominal dimensions 2.2 × 1.9 cm height were cut from each of the core samples.

Measurement of remanent magnetization was done using one of three commercially available magnetometers: a DIGICO slow speed (7 Hz) fluxgate spinner magnetometer or one of two 3.8 cm access 2-axis SQUID (Superconducting Quantum Interference Device) magnetometers inter-

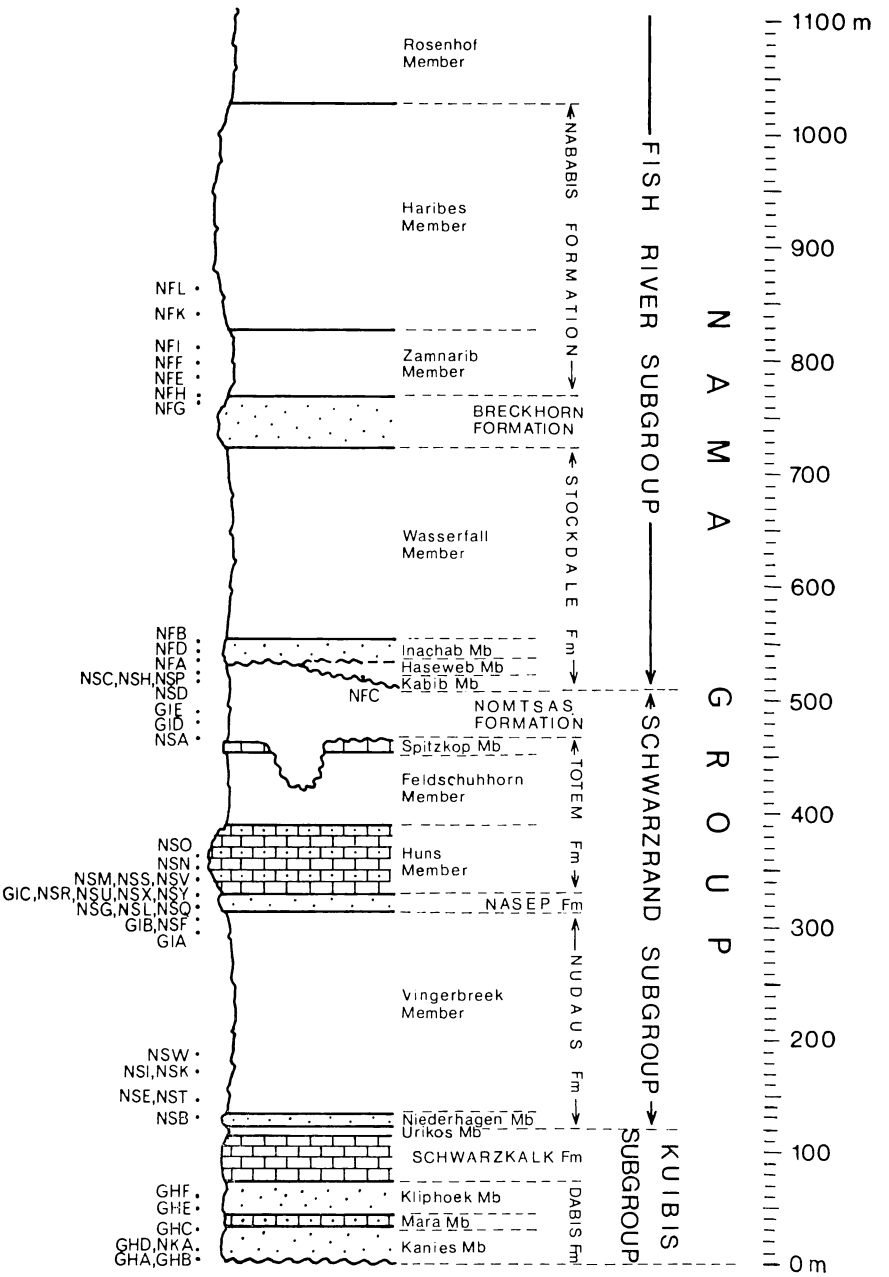


Fig. 3. Generalized section of the Nama Group, showing stratigraphic distribution of sampling sites.

faced to the DIGICO processor. A practical lower intensity limit of 2.5 $\times 10^{-4}$ A/m was realized for remanence measurements on the cryogenic instrument.

Thermal demagnetization experiments were done in air in a nonmagnetic furnace capable of obtaining temperatures in excess of 700°C surrounded by a 10 coil feedback-controlled field cancellation system. Residual DC magnetic field levels during cooling periods were kept below ±2 nT. Alternating field demagnetization was done with a 3-axis tumbler within a large solenoid capable of attaining peak fields of 210 mT. DC field cancellation within the AF apparatus was kept to within ±10 nT during demagnetization. Chemical leaching was performed in the same field controlled space used for thermal demagnetization. Specimens were serrated to facilitate more rapid acid penetration; in the initial 24 hr leaching period, HCl concentration was kept at 6N. Subsequent leaching was done in 12N HCl. Specimens were repeatedly rinsed in fresh water during a period of about 1 hr prior to measuring.

Transfer of partially demagnetized specimens from the particular demagnetization device to the magnetometer was done in mu-metal magnetic shields inside which DC magnetic fields were cancelled to within ±50 nT. Specimens were loaded in the magnetometer within Helmholtz coil sets which cancelled the ambient magnetic field in the sample handling region to ±100 nT.

Blaubeker Formation.—Samples of red quartzites from the lower Blaubeker Formation were collected at 8 sites in central Namibia (fig. 1). NRM site mean directions were loosely grouped with a general PEF trend, as shown in figure 4A. During thermal demagnetization, specimens from most sites exhibited poor directional stability but at high temperatures tended to group around a direction with a shallow to moderate positive inclination and northerly or southwesterly declination (one of the better examples is illustrated in fig. 5). Apart from PEF components subtracted during demagnetization, this was the only consistently observed magnetization direction.

Temperatures for bulk thermal treatment of remaining specimens were chosen by examination of stability ranges of individual pilot specimens; these ranges were generally 500° to 650°C. The final cleaned direction (after omission of one statistically random site mean direction) has a low overall precision (fig. 4B,C and table 2), although it is still significantly grouped at the 95 percent confidence level.

The distribution of cleaned site mean directions is clearly not Fisherian, either before or after structural correction. Two sparsely populated general groups are apparent: one defined by 8 samples from sites NBA and NBC, the other by 11 samples from sites NBE, NBF, and NBH. These latter sites yield a profoundly negative fold test (precision parameter ratio = 0.18, 95 percent significance point = 0.20) and are grouped to form the NBX mean direction. Samples from sites NBA and NBC come from a slightly different locality and lithology but have similar site mean directions which are quite different from the NBX direction. Results from

these two sites are difficult to interpret definitively, but these VGPs may reflect a pre-Damaran magnetization, as discussed in the following section. Sites NBD and NBG have reasonably well grouped mean directions which are considerably displaced from the other site mean directions.

Nama Group.—Nama Group samples of widely varying lithology were collected at 47 sites throughout southern Namibia (fig. 1). Four samples spanning approx 0.5 m of vertical section were collected at each site. Seven formations in the three subgroups were sampled, giving a good overall stratigraphic distribution of the sites (fig. 3).

Fish River Subgroup.—Site mean NRM directions from the Stockdale and Nababis Formations were loosely grouped and not significantly different from the PEF direction at the sampling site (fig. 6A). Pilot

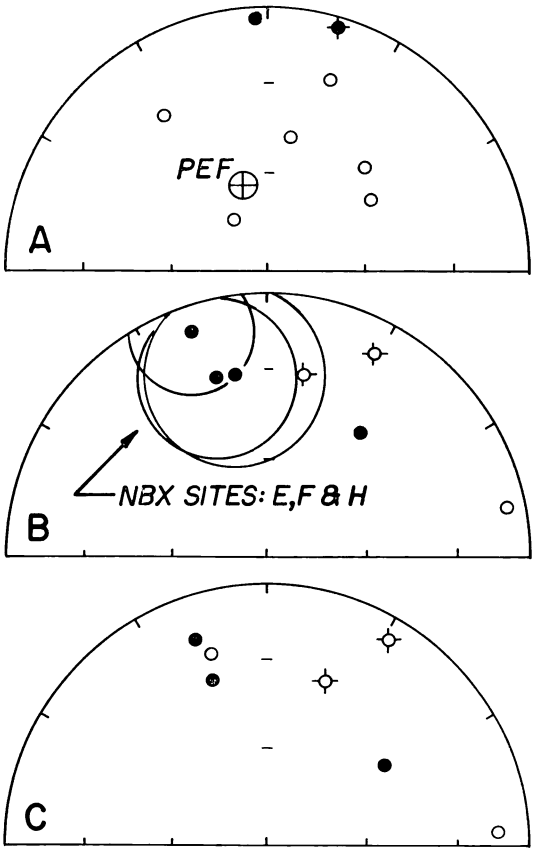


Fig. 4. Magnetization directions from Blaubeker Formation sites: (A) NRM directions, (B) after thermal demagnetization (*in situ*), (C) after thermal demagnetization (tectonically corrected). Open (filled) symbols represent negative (positive) inclination. Symbols with superimposed crosses refer to magnetization directions of opposite polarity which have been reflected through the origin to lie on the opposite side of the projection. Confidence circles (P=0.05) are shown about mean directions for NBX sites. Large PEF symbol—Present Earth's Field at the sampling site. Equal angle projection.

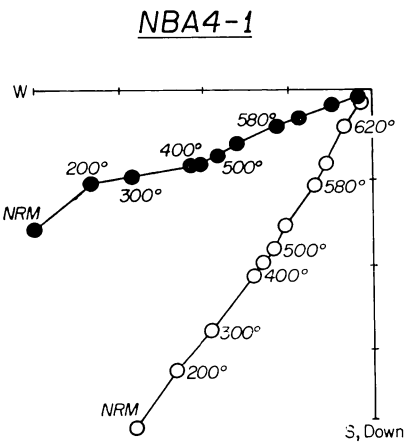


Fig. 5. Orthogonal vector plot illustrating progressive thermal demagnetization behavior of a specimen from the Blaubeker Formation. Open symbols, projection of vector endpoint upon horizontal plane; filled symbols, projection of vector endpoint upon north-south vertical plane. Axis ticks 1.0×10^{-8} A/m.

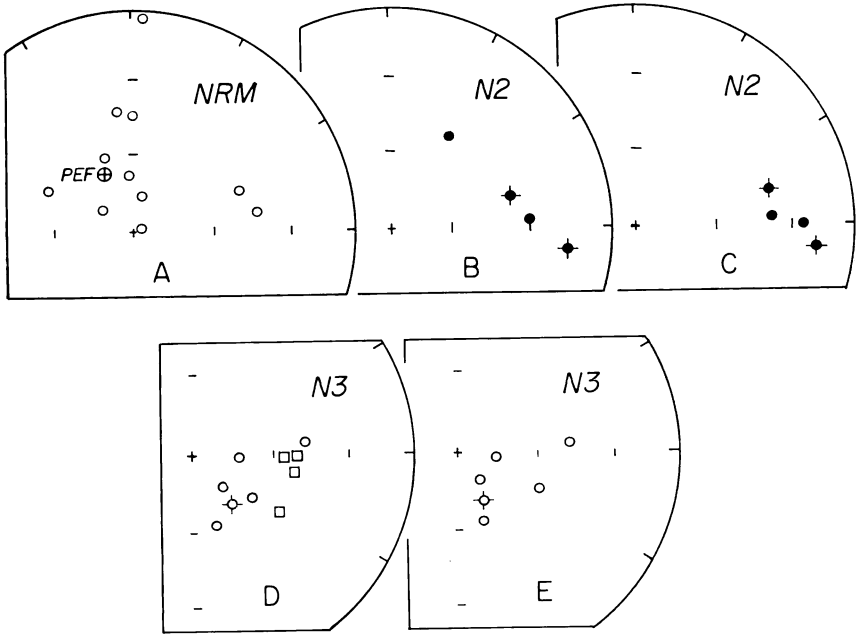


Fig. 6. Magnetization directions from Fish River Subgroup sites: (A) NRM directions, (B) N2 magnetization (*in situ*), (C) N2 magnetization (tectonically corrected), (D) N3 magnetization (*in situ*), (E) N3 magnetization (tectonically corrected). Squares in (D) are selected data from Piper, 1975; see text for details. Equal angle projection, conventions as in figure 4.

TABLE 2

A. Blaubeker Formation: site mean directions after thermal demagnetization

Site	D,I*	D,I**	(N)n	R	α_{95}	T
NBA	208,+14	211,+09	(4)3	2.69	33	600
NBB	191,+31	200,+33	(4)3	2.96	11	650
NBC	079,−08	087,−13	(4)3	2.88	20	650
NBE	349,+31	342,+33	(4)4	3.36	31	650
NBF	341,+10	344,−24	(3)3	2.86	22	500
NBG	036,+41	055,+45	(3)2	1.97	—	500
NBH	344,+29	341,+17	(4)4	3.61	25	650

Key: * Declination, inclination before tectonic correction; ** Declination, inclination after tectonic correction; (N): number of samples collected and measured; n: number used in final mean direction calculation; R: vector resultant, n unit vectors; α_{95}: semi-angle, cone of confidence about mean direction, P=0.05; T: Demagnetization temperature (°C).

B. Blaubeker Formation, overall summary

		(before tectonic correction)						(after tectonic correction)					
	N	D,I	R	k	α_{95}	λ,ϕ	dp,dm	D,I	R	k	α_{95}	λ,ϕ	dp,dm
All samples—unit weight to:													
Samples	22	011,+10	15.61	3	20	59,038	10,20	012,+02	14.54	3	23	63,043	12,23
Sites	7	014,+10	5.36	4	37	58,044	19,37	018,+02	4.92	3	43	60,054	23,43
NBX magnetization*—unit weight to:													
Samples	11	345,+24	9.69	8	18	51,352	10,19	342,+09	9.05	5	22	56,343	11,22
Sites	3	344,+23	2.95	45	19	51,353	11,20	342,+09	2.74	8	48	57,342	24,48

Key: N: number of samples or sites; D,I: Mean declination, inclination; R: vector resultant, N unit vectors; k: precision parameter; α_{95}: semi-angle, cone of confidence about mean direction, P=0.05; λ,ϕ: latitude and longitude of paleomagnetic pole; dp,dm: semi-minor and semi-major axes of polar error ellipse, P=0.05; * Sites NBE, NBF and NBH used in calculating NBX mean direction.

thermal demagnetization of at least one and often two specimens from each of 44 samples revealed several components of magnetization in these samples. Some pilot specimens had directions that remained aligned with the PEF direction during most of the 13 demagnetization steps, with the onset of directional instability occurring in the 660° to 670°C range. Other specimens underwent systematic directional changes during the lower temperature demagnetization steps, generally attaining directional stability in the 450° to 630°C range (fig. 7). Two groups of stable endpoints different from PEF directions were observed in samples from the Fish River Subgroup; these have been designated the N2 and N3 magnetizations in tables 3 and 4 and in figure 6. AF demagnetization of pilot specimens proved unsuccessful in removing completely a PEF component from samples that exhibited the N2 or N3 magnetizations, even when peak fields of up to 140 mT were employed. Thermal demagnetization was therefore used as the method of removing the (?) recently acquired component.

After bulk thermal treatment of remaining specimens, cleaned sample mean directions displayed either the N2 or N3 magnetization but never both at the same site. The N2 magnetization was observed in 15 samples at 4 sites and is characterized by an easterly (westerly) declination and a shallow to moderate positive (negative) inclination. The N3 component was observed in 14 samples at 4 different Fish River Subgroup sites and had an east-southeast (west-northwest) declination with a steep negative (positive) inclination. Excluding site NFI as discussed below, no consistent subtracted directions other than PEF vectors were observed in the pilot demagnetization studies. This would suggest that either the N2 or N3 magnetization (but not both) are present at any particular site, although overlapping blocking temperature spectra may have complicated the

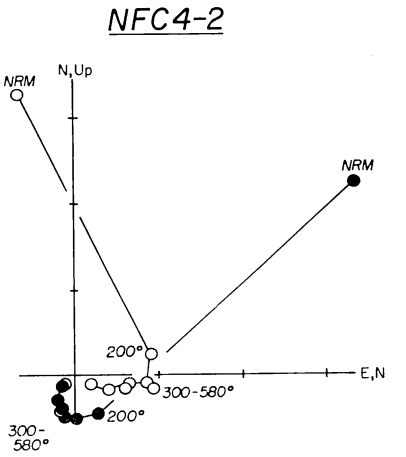


Fig. 7. Orthogonal vector plot illustrating progressive thermal demagnetization of specimen carrying N2 type magnetization from Fish River Subgroup. Conventions as in figure 5.

TABLE 3

Paleomagnetic results from the Nama Group. Symbols as in table 1

A. Fish River Subgroup: site mean directions after thermal demagnetization

	Site	(in situ) D,I	(corrected) D,I	(N) n	R	α_{95}	T°C	Magneti-zation
Stockdale Formation	NFA	321,+67	331,+70	(4)3	2.92	25°	500	N3
	NFC	087,+35	086,+39	(4)4	3.89	18°	350	N2
	NFD	348,–53	346,–51	(4)4	3.99	6°	500	PEF
Nababis Formation	NFE	256,–41	255,–39	(4)4	3.67	32°	550	N2
	NFF	162,–61	161,–62	(4)4	3.99	5°	500	N3
	NFG	095,–72	094,–76	(4)4	3.89	18°	640	N3
	NFH	277,–18	277,–19	(4)4	3.89	18°	500	N2
	NFI	105,–14	105,–15	(4)4	3.99	5°	600	?
	NFK	139,–73	141,–77	(4)3	2.84	37°	630	N3
	NFL	035,+48	089,+25	(4)3	2.95	20°	500	N2

B. Schwarzrand Subgroup: site mean directions after thermal demagnetization

	Site	(in situ) D,I	(corrected) D,I	(N) n	R	α_{95}	T°C	Magneti-zation
Nomtsas Formation	NSC	016,+17	017,+11	(4)4	3.84	22°	360	N1
	NSH	179,–12	181,–09	(4)4	3.55	38°	320	N1
	NSP	051,–01	050,+01	(4)4	3.80	25°	360	N1
	GID	239,–20	236,–22	(4)4	3.97	9°	320	N1
	GIE	085,–47	085,–47	(4)4	3.98	8°	500	N3
Totem Formation	NSM	171,+46	171,+46	(4)4	3.99	3°	450	N1
	NSN	126,–62	114,–57	(4)4	3.84	22°	360	N3
	NSO	023,–03	023,+02	(4)4	3.95	11°	360	N1
	NSV	193,–12	193,–07	(4)3	2.79	42°	360	N1
Nasep Formation	NSG	045,–23	041,–30	(4)3	2.99	9°	500	N1
	NSL	356,–47	354,–49	(4)4	3.99	5°	*	PEF
	NSQ	023,–05	019,–03	(4)4	3.81	24°	360	N1
	NSU	194,–12	193,–11	(4)3	3.87	20°	360	N1
	NSY	153,–28	151,+32	(4)2	1.97	—	*	N1
	GIB	285,–71	293,–68	(4)4	3.92	15°	500	?
	GIC	024,–28	025,–23	(4)4	3.80	25°	320	N1
Nudaus Formation	NSB	005,–34	002,–34	(4)4	3.78	26°	360	N1
	NSE	282,–18	282,–18	(4)4	3.76	27°	400	?
	NSF	039,–53	037,–54	(4)4	3.59	37°	400	PEF
	NSK	027,+16	031,+22	(4)4	3.65	33°	450	N1
	NST	023,–17	023,–10	(4)4	3.83	22°	360	N1
	NSW	002,–52	012,–53	(4)4	3.94	13°	*	PEF
	NSX	008,–32	006,–28	(4)4	3.99	4°	450	N1
	GIA	241,+21	241,+21	(4)4	3.90	17°	320	N1

* Chemical demagnetization

C. Kuibis Subgroup: site mean directions after thermal demagnetization

	Site	(in situ) D,I	(corrected) D,I	(N) n	R	α_{95}	T°C	Magneti-zation
Dabis Formation	NKA	021,–42	024,–45	(4)3	2.81	40°	400	N1
	GHA	174,–06	175,+13	(4)4	3.99	5°	500	N1
	GHF	200,+01	200,+01	(4)4	3.38	46°	500	N1

TABLE 3 (continued)

D. Nama Group: overall summary

Magnetization	n,N*	(in situ) D,I	R	k	α_{95}	λ,ϕ	dp,dm	(after tectonic correction) D,I	R	k	α_{95}	λ,ϕ	dp,dm
N1 component:													
Nomtsas Fmn	16,4	031,+13	3.62	8	35°	45,064	18,36	031,+12	3.67	9	32°	46,065	17,33
Totem Fmn	11,3	011,–12	2.67	6	55°	67,046	29,56	011,–12	2.68	6	54°	67,046	28,55
Nasep Fmn	17,5	017,–04	4.30	6	35°	60,054	18,35	015,–17	4.46	7	30°	67,059	16,31
Nudaus Fmn	20,5	025,–19	4.51	8	29°	61,077	15,30	025,–16	4.42	7	31°	60,074	17,32
Dabis Fmn	11,3	191,+12	2.73	7	49°	67,046	25,49	192,+20	2.78	9	43°	70,054	24,45
N1 overall:													
unit weight sites	(20)	020,–07	17.32	7	13°	60,061	07,13	020,–11	17.51	8	13°	61,063	06,13
unit weight fmns	(5)	019,–07	4.86	29	14°	61,059	07,14	019,–11	4.85	27	15°	62,061	08,15
N2 component:													
Stockdale Fmn	4,1	087,+35	—	—	—	06,269	—	086,+39	—	—	—	06,266	—
Nababis Fmn	11,3	253,–39	2.75	8	47°	–04,260	33,56	268,–28	2.94	33	22°	05,273	13,24
N2 overall:													
unit weight sites	(4)	077,+38	3.73	11	29°	–01,263	20,34	087,+31	3.93	41	15°	05,271	09,16
unit weight fmns	(2)	080,+37	1.99	—	—	01,265	—	087,+34	1.99	—	—	06,269	—
N3 component:													
Nababis Fmn	3,1	321,+67	—	—	—	07,353	—	331,+70	—	—	—	06,001	—
Stockdale Fmn	11,3	137,–71	2.94	35	21°	00,355	32,37	140,–74	2.95	38	20°	–02,359	33,36
Nomtsas Fmn	4,1	085,–47	—	—	—	–16,311	—	085,–47	—	—	—	–16,311	—
Totem Fmn	4,1	126,–62	—	—	—	05,341	—	114,–57	—	—	—	01,331	—
N3 overall:													
unit weight sites	(6)	122,–67	5.80	25	14°	–02,344	19,23	120,–68	5.76	21	15°	–04,344	21,25
unit weight fmns	(4)	116,–64	3.87	24	19°	–03,339	24,30	114,–64	3.85	20	21°	–04,338	27,34

* n: number of samples, N: number of sites.

analysis. A direct comparison of the relative blocking temperatures of the N2 and N3 components was therefore not possible. However, a fold test (table 5) indicates that the N2 component was probably acquired prior to folding of the Fish River Subgroup during the early Paleozoic Damaran event, while the N3 magnetization was possibly acquired after folding. A magnetization age sequence N2 > N3 is indicated.

AF demagnetization experiments conducted on a number of pilot specimens showed that a very high coercive force is exhibited by the magnetic minerals that carry the PEF and N3 components and that thermal demagnetization was a much more suitable technique for isolating the N2 and N3 magnetizations. With this in mind, a reexamination of the paleomagnetic data presented in Piper (1975) suggests that the apparently streaked distribution of sample and site mean directions may be due to incomplete removal of a PEF component coexisting with an N3-type component in some sites. The non-PEF endpoint of the distribution of Piper's data is similar to the N3 direction observed in the Fish River samples described above. By rejecting 4 sites (95,96,97,100) whose mean directions are not significantly different from the PEF at the sampling site and a fifth with an indeterminate mean direction (94), the remaining 4 sites (squares in fig. 6D) define a mean direction similar to the N3 mean direc-

TABLE 4

A. Fish River Subgroup: data of Piper (1975).
AF demagnetization to 100 mT, no structural correction

Site	n	D	I	α_{95}	Magnetization
94	5	002	−16	07	?
95	5	008	−54	24	PEF
96	6	037	−52	12	PEF
97	5	350	−37	21	PEF
98	4	101	−51	24	N3
99	6	091	−52	13	N3
100	3	002	−49	33	PEF
101	6	091	−56	38	N3
102	4	125	−51	20	N3

B. N3 site mean directions and pole positions; no structural correction

	D,I	S	R	k	α_{95}	λ,ϕ	dp,dm
1. N3 sites (Table 3D)	122,−67	6	5.80	25	14	−02,344	19,23
2. N3 sites (Table 4A)	102,−53	4	3.95	65	11	−05,323	11,15
3. overall	112,−62	10	9.65	26	10	−04,335	12,16

(1) and (2) above are not significantly different (P=0.05): F=3.20, 95% significance point = 3.63.

Key:

n: Number of samples
D,I: Mean declination, inclination
α_{95}: Semi-angle, cone of confidence about mean direction (P=0.05)
S: Number of sites
R: Resultant vector length
k: Precision parameter
λ,ϕ: Latitude, longitude of paleomagnetic pole
dp,dm: Semi-minor, semi-major axes of polar error ellipse (P=0.05)

tion (table 4). An F-ratio test indicates that the structurally uncorrected N3 mean direction is not significantly different (P=0.05) from the mean direction defined by Piper's sites 98, 99, 101, and 102. N3-type magnetizations are found at two sites in the underlying Schwarzrand Subgroup, and therefore it seems possible that the N3 component may be the result of a post-diagenetic overprinting of preexisting primary (?) N2 magnetizations in the Fish River Subgroup. If this is indeed true, the paleomagnetic results presented in Piper (1975) probably have more relevance to secondary overprinting than to an original primary magnetization.

The NFI site mean direction (table 3) is problematic, as it is statistically distinct from both the N2 and N3 mean directions, as well as the PEF direction. The site mean direction lies near the great circle segment joining the N2 and N3 mean directions, and thus two possible explanations are apparent. Either the NFI site mean direction represents an intermediate age magnetization acquired in the time interval between times of acquisition of the N2 and N3 components, or coexisting N2 and N3 magnetizations are present in the NFI samples and have not been separated during demagnetization. Unfortunately, directional instability problems caused by the production of new high susceptibility magnetic phases at elevated temperatures during thermal demagnetization prevented more detailed investigation. Considering the exceptionally stable directional behavior during the lower (20°-650°C) temperature demagnetization steps and the tight, non-streaked grouping of the individual specimen directions from site NFI, the former interpretation is favored, and the NFI site mean pole is tentatively regarded as a VGP (Virtual Geomagnetic Pole) intermediate in age between the N2 and N3 poles (fig. 10).

Schwarzrand and Kuibis Subgroups.—Site mean NRM directions from the 5 formations sampled in the Schwarzrand and Kuibis Subgroups were scattered, with a general trend toward the PEF direction (fig. 8A). Pilot thermal demagnetization studies were conducted on at least one and often two specimens from each of 144 samples. Variable behavior of the magnetization vector during pilot thermal and chemical demagnetization was noted. Three sites exhibited very stable PEF magnetizations, while two others reached stable endpoints essentially identical to the N3 magnetization described above. These latter sites have been grouped with the Fish River Subgroup N3 sites to form the N3 mean direction in tables 3 and 4. In two sites (NSE,GIB), well-grouped cleaned mean directions were

TABLE 5

Fold test: N1, N2, N3 magnetizations

Component	S	k'/k	Significance point* 90%	95%	99%
N1	20	1.07	1.51	1.70	2.12
N2	4	3.62	2.78	3.79	6.99
N3	6	0.84	0.45	0.35	0.22

S: number of sites

k'/k: precision parameter ratio; k (k') before (after) tectonic correction

* F-ratio at equal 2(S-1) degrees of freedom

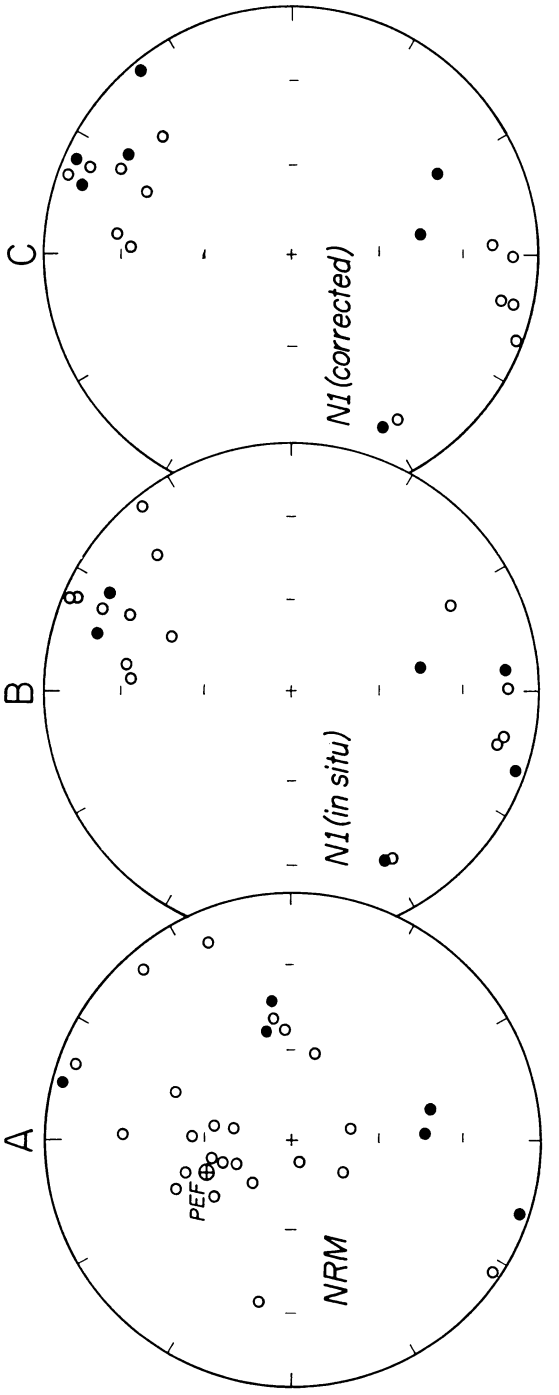


Fig. 8. Magnetization directions from Schwarzrand and Kuibis Subgroup sites: (A) NRM directions, (B) N1 magnetization (*in situ*), (C) N1 magnetization (tectonically corrected). Equal angle projection, conventions as in figure 4.

observed which did not appear to be related to any of the other observed magnetizations. These site mean directions have not been included in any of the overall mean magnetization directions.

In the majority of sites, a significant PEF component was removed in the initial heating steps, with remanence vectors stabilizing to a shallow north-northeast (south-southwest) direction, generally in the 350° to 500°C range (fig. 9). This stable magnetization has been designated the N1 component and was observed in 75 samples from 20 sites in 5 formations (table 3 and fig. 8). Chemical demagnetization experiments were performed on specimens from 3 sites with a dominant red lithology (NSL, NSW, NSY). Sites NSL and NSW exhibited a very stable PEF-type magnetization, while an N1-type magnetization was observed at site NSY. Directional behavior of the magnetization vector at this latter site during chemical demagnetization was stable and essentially identical to directional behavior during thermal demagnetization; no new information was gained in the acid leaching experiments. Thermal demagnetization was chosen as the routine demagnetization method for final treatment of remaining specimens from each sample. An F-ratio test performed on the N1 component (table 6) indicates that the final thermally cleaned mean directions from each of the 5 formations are not significantly different at the 95 percent confidence level, and therefore they have been grouped to define the overall N1 mean direction.

Upon tectonic correction, overall precision of the N1 magnetization increases slightly when unit weight is given to site mean directions (table 5). The change is not significant at the 95 percent confidence level due to the everywhere shallow dip of the beds. The fold test is only suggestive of pre-folding magnetization age; however, there are three additional lines

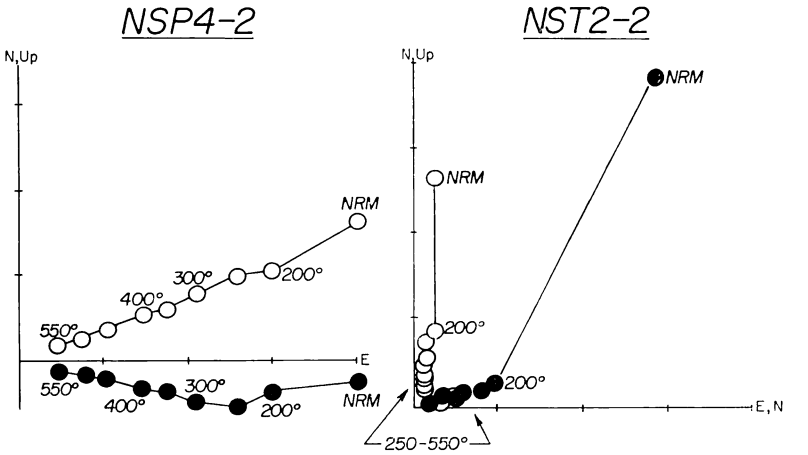


Fig. 9. Orthogonal vector plot illustrating progressive thermal demagnetization of specimens carrying N1-type magnetizations from Schwarzrand Subgroup. Conventions as in figure 5, except that filled symbols here are projection of vector endpoint upon east-west vertical plane.

of evidence that can be cited in support of a primary (penecontemporaneous with deposition) magnetization age for the N1 component:

1. Antipolar normal and reversed magnetizations are present at the same locality in sites separated by as little as 15 to 20 m in stratigraphic section within each of the 5 formations and occasionally within an individual site. Although the inferred overprint component N3 also exhibits both polarities, positive inclination for this component is observed only at site NFA, which is separated from the other N3 sites by a considerable lateral distance and thus may have an entirely different thermal and chemical history.

2. An early Paleozoic (?Damaran) overprint (N3) with a negative fold test and different in direction from N1 occurs throughout the entire Nama Group. If N1 were an overprint component, it might be expected that N1-type magnetizations might be found in the unconformably overlying Fish River Subgroup as well as in the Schwarzrand and Kuibis Subgroups. This is contrary to the stratigraphic distribution of the N1 component, which is observed only in the lower Nama Group.

3. The N1 pole position is different from the later Phanerozoic APWP for Africa, although its antipole is near (but significantly removed from) Cambro-Ordovician poles.

Two options are apparent: either the N1 magnetization is primary and of depositional/diagenetic age, or it is secondary and of a general early Paleozoic (that is, Damaran metamorphism) age. The N3 magnetization is observed in the Fish River, Schwarzrand, and Kuibis Subgroups; its corresponding pole lies in a post-N2 position on the proposed APWP for southern Africa (fig. 10), consistent with the hypothesis that the N3 magnetization has totally overprinted both the N1 and N2 components in parts of the Nama Group. Considering the available evidence, the N1 magnetization is interpreted as predating the N3 component and therefore is probably of depositional or diagenetic age. The magnetization age sequence N1 > N2 > N3 is therefore favored for the three magnetizations isolated in the Nama Group.

MAGNETIZATION AGES: N1, N2, N3, NBX

Paleomagnetic pole positions calculated from the Blaubeker Formation and the Nama Group are plotted in figure 10 together with previously published late Precambrian and early Paleozoic data from southern Africa (table 7). We take the Cambrian-Precambrian boundary to be 560 m.y., following the recommendation of IGCP Project 29 (J. Cowie,

TABLE 6

Comparison of formation mean directions, N1 component. Nomtsas, Totem, Nasep, Nudaus and Dabis Formations

S	N	R	ΣR_i	ΣN_i-R_i	F
5	20	17.51	18.01	1.99	0.94 (95% significance point = 2.38)

personal commun., 1978). Magnetization age for the NR pole from the Ntonya ring structure is interpreted as more closely related to biotite K/Ar ages of 474 and 479 m.y. than to the 630 m.y. Rb/Sr age associated with this pole previously. As suggested in a previous section, the N3 magnetization is probably of post-folding age. The Damaran episode was the most recent tectonothermal event to have affected the sampling area, and the location of the N3 pole on the African APWP is entirely consistent with an early Paleozoic magnetization age, by comparison with the available data.

The N2 pole predates the Damaran episode, to judge by the positive fold test which is significant at the 99 percent confidence level. By com-

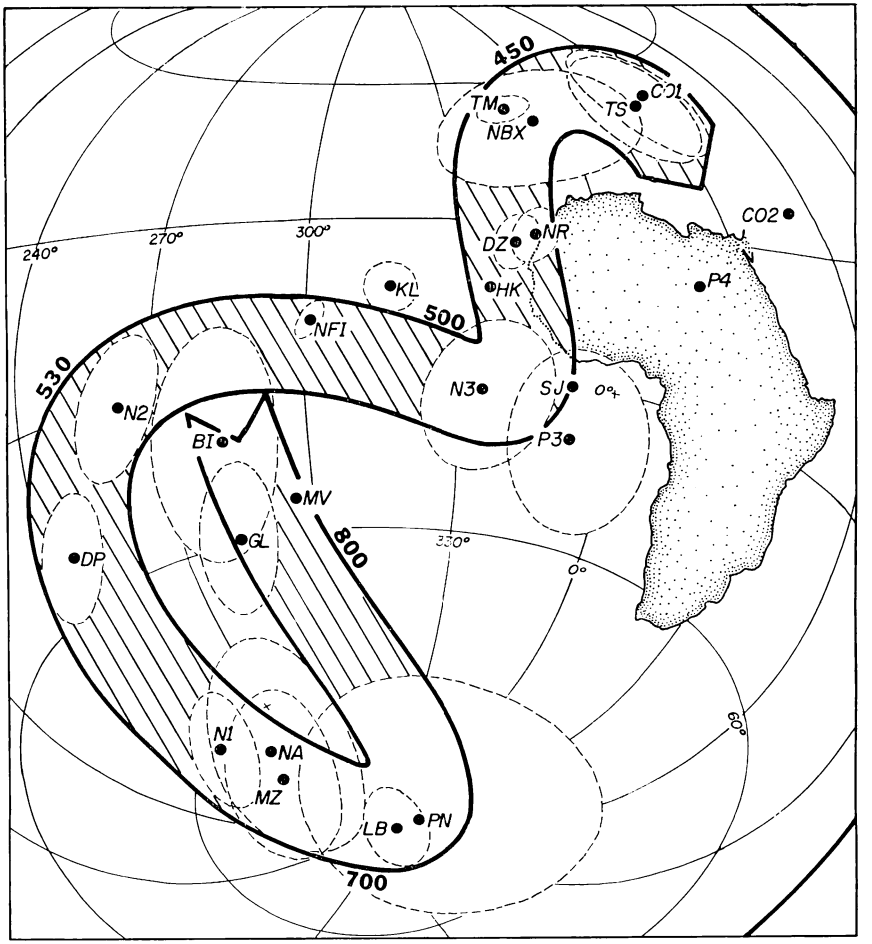


Fig. 10. Proposed 450 to 800 m.y. southern APWP for Africa. Dashed ellipses are polar uncertainties dp,dm (P=0.05). Symbols as in table 7. Error ellipses not plotted for poles with large uncertainty. Africa is fixed in its present position; grid lines at 30 degree intervals for latitude and longitude. Swathe width 15 degrees at equator; equal area projection.

parison with other African data, the N2 pole lies in a position intermediate in age between early Paleozoic poles and poles from the underlying parts of the Nama Group, (?) Blaubeker Formation, and Mbozi Complex of Tanzania. New paleomagnetic data from the lower Mulden Group in northern Namibia (McWilliams and Kröner, ms) indicate that the magnetization in these rocks was also acquired before Damaran folding; the pole position from the Mulden Group is statistically indistinguishable from the N2 pole from the Nama Group. Isotopic studies indicate a maximum depositional age of 565 to 570 m.y. for the Mulden samples, with a preferred age of approx 530 to 540 m.y. (Kröner and Clauer, 1979), and a similar older age constraint can be placed on the N2 magnetization. A younger age limit for N2 is given by the better defined Cambrian and Ordovician poles on the APWP which lie on a later segment of the APWP. Thus an early Cambrian age for deposition of the Fish River Subgroup and the N2 magnetization is suggested. This is in excellent agreement with independent geological evidence which suggests a possible Cambrian age for the Fish River Subgroup (Germs, 1972a).

TABLE 7

Paleomagnetic data for southern Africa, 400 to 800 m.y.

	Rock Unit	Magnetization age	Pole	Reference
P4	Plateau Series	PA2 (?)	60N 025E	G14 (560)
CO1	Groupe de la Falaise d'Atar	Є - O	36N 052E	01
CO2	Groupe des Plateaux d'Oujeft	Є - O	55N 030E	01
TS	Tassili sediments	Є - O	53N 026E	02
TM	Table Mountain series	O	50N 349E	G04 (032)
NBX	Blaubeker Formation	471±26	51N 352E	this study
NR	Ntonya ring structure	474,479 K	27N 345E	G09 (137)
DZ	Dedza Mt. syenite (VGP)	≅ NR	27N 340E	G14 (150)
HK	Hook intrusives	500±17 R	14N 336E	G09 (132)
SJ	Sijarira Group	Є - Pc	02N 352E	G12 (149)
P3	Plateau Series	PA2 (?)	10S 352E	G14 (562)
N3	Nama Group, overprint	PA2 (?), < N2	02S 344E	this study
KL	Klipheuval Formation	Є - Pc	16N 316E	03
NFI	Nama Group, site NFI (VGP only)		10N 301E	this study
N2	Nama Group, Fish River Subgroup	< N1	05N 271E	this study
DP	Doornpoort Formation	500 - 550	22S 245E	G14 (529)
N1	Nama Group, Schwarzrand & Kuibis	> 553±13, R, < 686±32 R	62S 241E	this study
NA	Blaubeker Formation, sites NBA, NBC	PA1 (?)	62S 264E	this study
MZ	Mbozi Complex	743±30 K	72S 248E	G14 (539)
LB	Lower Buanji Group		87S 083E	G14 (538)
PN	Pre-Nama dykes	653±70 K (?)	85S 048E	G14 (530)
GL	Gagwe lavas	813±30 K	29S 283E	G14 (540)
MV	Manyovu redbeds		22S 297E	G14 (545)
BI	Bukoban intrusives	806±30 K	11S 281E	G14 (541)

References: G prefixes refer to Geophysical Journal pole lists, with the format G [list no.] (pole No.); 01—Morris and Carmichael (1978); 02—Ileana (1971), in Piper (1976); 03—Creer (1973).

Ages: K—K/Ar determination; R—Rb/Sr determination. PA1 and PA2 refer to early and late Pan-African metamorphic events (see text for details).

The N1 pole lies in a position between dated poles GL, BI, PN, MZ, and younger poles DP and N2. Apart from the evidence for original magnetization presented in the section entitled Paleomagnetic Results, the age constraints on these poles are consistent with independent field evidence presented in Regional distribution, stratigraphy, and age (p. 945-947) which constrains the age of the Nama. Additional support for a primary rather than secondary magnetization age for N1 comes from the excellent agreement of directions from virtually unmetamorphosed and slightly metamorphosed sites (boundary in fig. 1). If Damaran metamorphism had affected preexisting magnetizations at sites in the epimetamorphic zone, directions from these sites should differ from directions from sites in the anchimetamorphic zone. This is not observed, and thus the N1 magnetization in each of the two zones must be of a very similar age. Mineralogic studies of clay minerals from Kuibis Subgroup rocks in the anchimetamorphic zone show that Damaran metamorphic effects were minimal in this zone and indicate a late Precambrian depositional age for the lower Nama Group. Based on these considerations and the independent paleontological data of Germs (1972a) and Pflug (1974 and personal commun., 1978) we infer an age of about 630 to 650 m.y. for the N1 magnetization, in good agreement with field evidence and position of the N1 pole with respect to other African paleomagnetic data.

The NBX pole from the Blaubeker Formation sites NBE, NBF, and NBH is almost certainly not primary to judge by the negative fold test and is not significantly different from the TM pole of Ordovician age. This agrees well with the previously mentioned isotopic data which indicate a postmetamorphic cooling age of about 470 m.y. in the nearby Damara belt; this is probably the magnetization age of the NBX pole. Directions from sites NBA and NBC (combined to form the NA pole of table 7) are different from the NBX sites and lie near the N1 pole from the lower Nama Group. A late Precambrian, early Pan-African (580-680 m.y.) magnetization age for these sites is possible, but more work will be required to enable a definite conclusion to be made.

CAMBRIAN-PRECAMBRIAN BOUNDARY

As discussed above, the N2 magnetization and therefore the Fish River Subgroup are most probably Cambrian in age, while the Schwarzrand and Kuibis Subgroups are probably Precambrian. Thus in this interpretation, the Cambrian-Precambrian boundary in the Nama Group may be represented by the unconformity that separates the Nomtsas Formation from the Stockdale Formation (fig. 3). However, there appears to be a definite paleontological break between the Nomtsas Formation and the underlying parts of the Nama Group; the earliest appearance of the trace fossil *Phycodes pedum* (thought to be globally diagnostic of a Cambrian age) occurs in the Nomtsas Formation, and this has led to the suggestion that the locally unconformable Nomtsas-Totem interface may be the Cambrian-Precambrian boundary in the Nama Group (Daily, 1973). Since the N1 pole from the Nomtsas is not significantly different

from the N1 pole from the underlying Nama units, there are two immediate alternatives:

A. If the N1 magnetization is not primary but is in fact of Paleozoic age, the Cambrian-Precambrian boundary may be within or below the Nomtsas Formation as suggested by the presence of *Phycodes pedum,* or

B. If N1 is indeed primary or at least late Precambrian in age, *Phycodes pedum* may not be restricted to the Cambrian.

The paleomagnetic data clearly suggest a time break between the Fish River Subgroup and the Schwarzrand Subgroup, rather than between the Nomtsas Formation and Totem Formation. The stratigraphy also suggests such a hiatus, since the youngest unit of the Schwarzrand Subgroup is only preserved north of Maltahohe. Farther south it was probably eroded prior to deposition of the Fish River beds. In view of the strong arguments for a primary magnetization age for N1, the second alternative is tenable, with the obvious conclusion that *Phycodes pedum* may not be suitable as a trace fossil for intercontinental chronostratigraphic correlation.

A possible reconciliation of (A) and (B) can be accomplished by postulating a number of somewhat special conditions. If N1 is primary and *Phycodes pedum* is a valid Cambrian indicator, then N2 must be younger than earliest Cambrian, which would be represented by N1. A middle Cambrian age for N2 would not be unreasonable, considering its position with respect to other Cambrian poles. Further, the directional similarity of N1 from the Nomtsas and N1 from the underlying Nama may be explained by adopting a quasi-static interval (little or no apparent polar motion) for the approximate period 560 to 650 m.y. Thus N1 would span the Cambrian-Precambrian boundary, while N2 from the Fish River Subgroup could be as much as 30 to 50 m.y. younger. A period of rapid apparent polar motion would be required to account for the 71 degree difference between the N1 and N2 poles; however, this polar shift is predominantly a paleoazimuthal change and requires only 19 degrees of latitudinal motion. This would correspond to true lateral displacement of the continental lithosphere at an average rate of 4 to 7 cm/yr, depending upon the age difference between N1 and N2.

GLACIAL PALEOLATITUDES

As discussed in the introductory section, there is evidence to suggest that glacial or glacio-marine sedimentation (the Klein Karas glacials) occurred during deposition of the Nama Group. The structurally corrected N1 mean direction from the Schwarzrand and Kuibis Subgroups has a mean inclination of 11 ± 12 degrees. This field inclination range would correspond to depositional paleolatitudes ranging from 0 to 12 degrees. Comprehensive discussion and analysis of the latitudinal distribution of late Precambrian glacial deposits and theories of glaciation are deferred until new results from elsewhere in Africa, Australia,

and North America have been presented; however several points should be noted.

If a conventional (that is, high latitude, circumpolar only) model for glaciation is adopted, the low latitude Klein Karas glaciation must have occurred as a restricted mountain glaciation similar to those presently observed in equatorial regions of east Africa (Ruwenzori Mts. and Mt. Kilimanjaro) and South America (Andean mountain chain). However, there is no evidence that such mountains existed in southern Namibia during deposition of the Schwarzrand Subgroup; indeed the Schwarzrand lies conformably almost everywhere on the Kuibis Subgroup, and the middle Schwarzrand Huns limestone contains marine stromatolites.

Although some of the glacial features associated with the Schwarzrand strata provide a clear indication of the presence of ice, it is not possible to relate them to a grounded ice sheet. Martin (1965) interpreted the grooved and striated surfaces to be produced by floating and sometimes grounded ice floes, and Kröner and Rankama (1973) suggested that the deep and long channels in some areas were not carved and filled by moving ice but by strong river currents in a wet and probably cold climate. This leaves the possibility of an areally extensive glaciation elsewhere in Gondwana at around 600 to 650 m.y. in order to account for localized glacial marine conditions at or near sealevel in equatorial latitudes during Schwarzrand times.

The best candidate for this is the widespread and undoubtedly continental glaciation in the Taoudeni basin of West Africa (Trompette, 1973) which has an age of about 610 to 650 m.y. (Clauer, 1976; Trompette, 1973). If the N1 magnetization dates from approximately this time, and the West African craton was in its present position with respect to southern Africa, this glaciation occurred in paleolatitudes of about 45 degrees, an acceptable position in comparison with glaciated terraines of the last circumpolar Quaternary ice age. Nevertheless, world climatic conditions must have been severe to allow ice to drift into equatorial latitudes, and we suspect that the circumpolar model alone is not sufficient to explain the glacial phenomena observed in Nama Group sediments.

One of the characteristics of most late Precambrian glaciogenic strata is their association with stromatolite-bearing carbonate rocks (Schermerhorn, 1974), and the Schwarzrand Subgroup is no exception (table 1). The lithological features observed in both the Schwarzrand and Fish River beds indicate pronounced climatic seasonality in equatorial latitudes, a phenomenon that is difficult to reconcile with the circumpolar glaciation model, unless drastic changes in the global circulation pattern of ocean currents and perhaps in the arrangement of continental masses are inferred (Crowell and Frakes, 1970). In view of the uncertainty about the relative positions of the continents during the late Precambrian, particularly as regards the various Gondwana crustal segments, we see no reason at this stage to reject either the circumpolar or the low latitude glaciation model of Williams (1975) on the basis of the Nama data.

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